

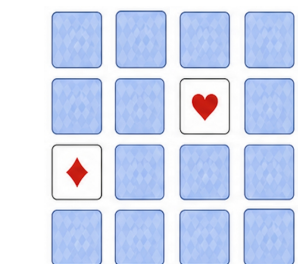
RNGBench: Two Non-Markov Game Settings

Both environments isolate memory for action under controlled settings.



Matching Pairs

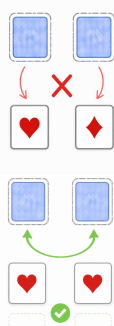
Static, Discrete Hidden State



Current Revealed Card Identities (And Pos)

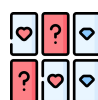


Action

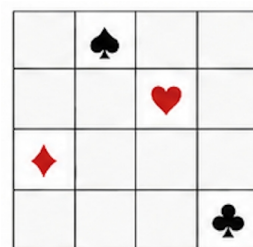


One Turn
Two Flip:
Not matching,
Automatic
Flip Back.

Matching,
They will be
removed
from board.



Hidden
State



Locations and Identities
of Observed Pairs



Optimal
Play

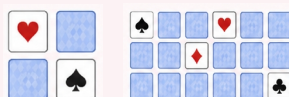
- ✓ Exploit known pairs.
- ✓ Explore Only when no match is known.
- ✓ Avoid redundant re-flips.

Pattern Variants



Noise Poker Texture

Size Variants



Length and width
change arbitrarily



Duel Mode

Same Board, Share Observation
One Fail Another Chance!



P1



P2



Eval Metrics



Score (%)

Fraction of
matched Pairs

Resp / Score

Responses per
matched pair



Duel Win(%) / Elo

Special In Duel Mode,
Pairwise win-based rating

PF(%) / IA(%)

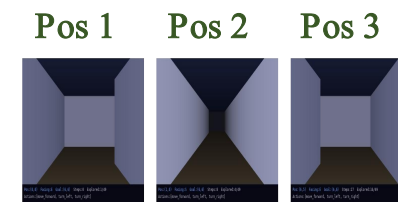
Action Parse Failure
Invalid Action

3 D Maze

Dynamic, Spatial Hidden State

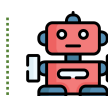


Observation



Have Ego-View of Maze

Different Position,
Different orientations,
Different View

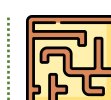
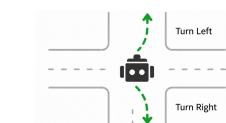


Action

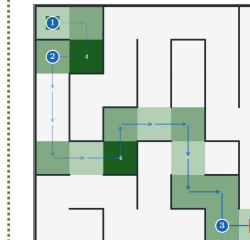
1) Move: Forward
Until Intersection



2) Turn: Choose
Turn Left Or Right



Hidden
State



Global maze layout,
visited cells, position,
and orientation



Optimal
Play

- ✓ Reconstruct map and pose.
- ✓ Avoid revisits and wall hits.
- ✓ Move toward goal or unexplored frontier.

Pattern Variants



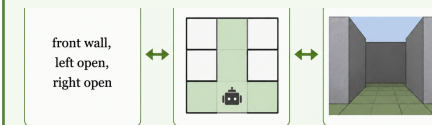
Poster Color Texture Plain

External Memory



No Map: Internal Memory.
MiniMap: See visited Path.

Modality



Text 2D Patch 3D View



Eval Metrics



Game Score

Measures success,
efficiency, and exploration

Eff(%)

Optimal path
/ actual path

Explore(%)

Ratio of maze
cells visited



Wall Hits(%)

Wall collisions